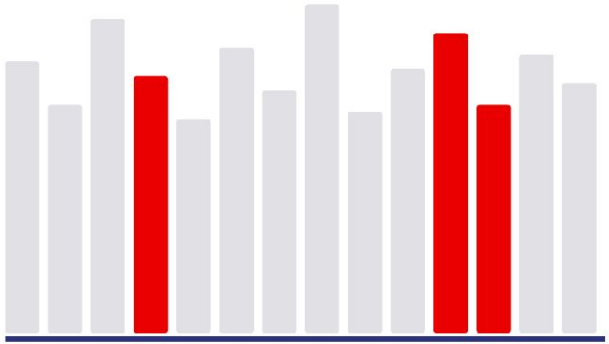
AI LITERACY SERIES: FROM THEORY TO PROTOTYPE

I built my own PM Copilot

An AI study buddy that answers only from the notes I trust.



14 PM guides. The red ones hold the answer.

Riya Murdia

THE PROBLEM

Interview in 2 days. Where do I revise?

General AI tools

Great answers, but long, generic, and not what I actually studied.

Too generic

My 14 PM PDFs

Exactly what I studied, but finding one concept means scrolling page by page.

Too slow

What I needed

Short answers from my own notes, with the page to check when I want depth.

So I built this

THE USE CASE

Written as a job to be done

“When I have an interview coming up, I want to brush up on PM concepts from sources I trust, so I can revise fast and walk in confident.”

What success looks like

Fast
Short, focused answers

Grounded
Only from my 14 documents

Verifiable
Points me back to the source

THE SOLUTION

A copilot built on my own notes

Ask in plain words

Type a question like you'd ask a friend

Answers from my notes

Uses only the pages that match best

Honest when it can't

Says “not covered” instead of guessing

Example: the two kinds of answers

What is TAM?

TAM is the total demand for a product if you won every possible customer.

Source: TAM SAM SOM MasterClass, p.3

What is the CEO's phone number?

Not covered in the provided documents.

LIVE DEMO

Let's see it run

Live on my laptop. No internet AI, no API key.

1 **A question my notes cover**

“What's the difference between TAM, SAM and SOM?”

You'll see: a short answer from my PDFs

2 **A question my notes don't cover**

“What is the CEO's phone number?”

You'll see: “Not covered in the provided documents.”

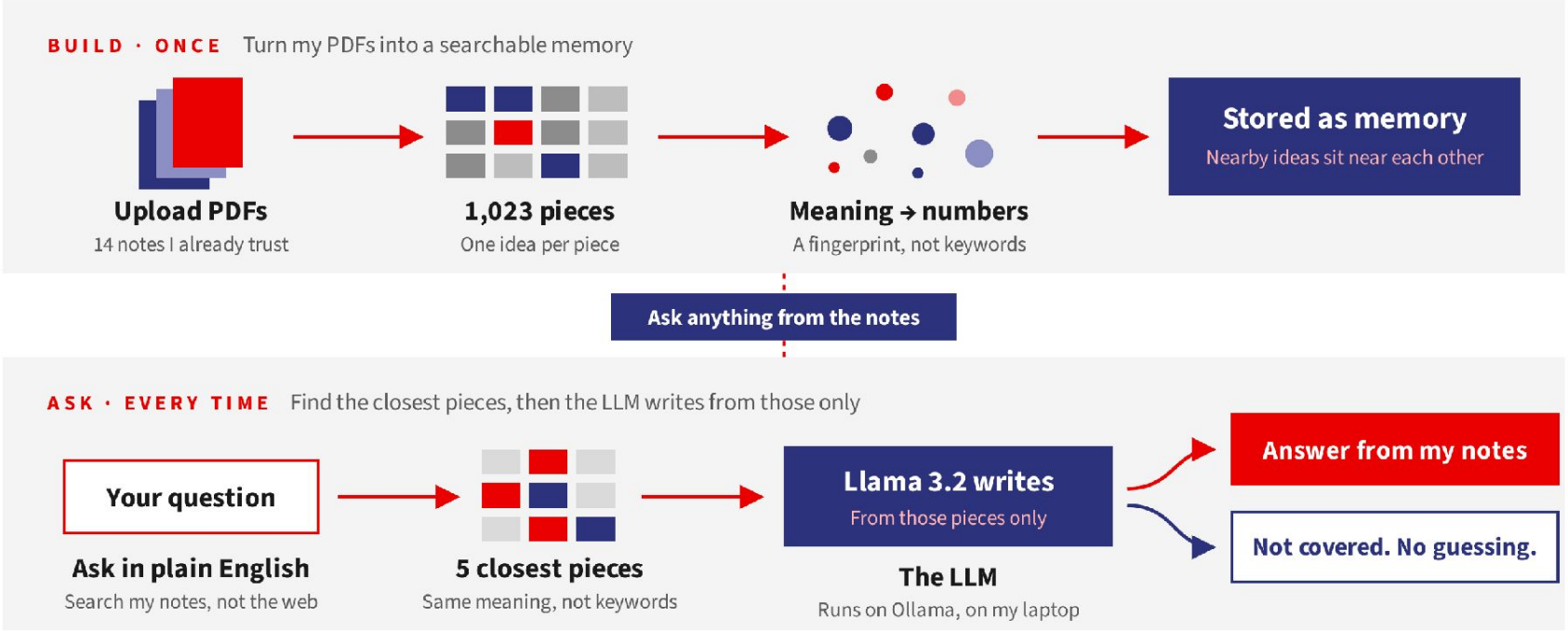
Built with Python, ChromaDB, MiniLM embeddings, Llama 3.2 on Ollama, and Streamlit

▶ **Watch the demo video**

HOW IT WORKS

Build the memory once. Ask it forever.

Answers come from my notes, not the internet. In AI terms, this is called RAG.



HOW I BUILT IT

The numbers I chose, and why

1,023

pieces (chunks)

Cut from 14 PDFs, so each piece holds one idea

800

characters per piece

About a paragraph, so search stays precise

150

characters of overlap

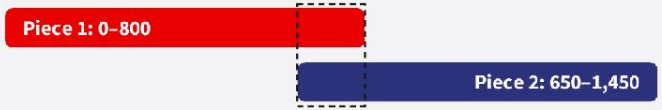
So no sentence gets cut in half

Top 5

pieces sent to the AI

Enough evidence, without extra noise

How overlap works



The 150 shared characters keep sentences whole.

Why free models on my laptop?

Private and $0 to run. The trade-off: small models are weaker at citing sources.

WHAT I KEPT IN MIND AS A PM

The guardrails that keep answers trustworthy

The AI model is only the writer. Guardrails decide what it may use, when it must refuse, and how I check it.

Grounded answers

It uses only pieces from my PDFs. No internet, no generic AI knowledge.

Honest refusal

If my notes don't cover it, it says so. A confident guess is a failure.

Private and free

Everything runs on my laptop. My notes stay private, and it costs $0.

Traceable

Every question and answer is logged, so I can check what happened.

HOW I CHECK IT

Questions I already knew the answer to.

Refusal

When my notes don't have it, does it refuse, or invent?

Accuracy

When it answers, is it from the right notes, with a real source?

EVALUATION

How I test every answer

I wrote 14 test questions and labelled them myself: 8 my notes can answer, 6 they can't.

1

Did it refuse at the right time?

Answer what my notes cover. Say “not covered” for the rest.

Yes No

2

Is the answer true to my notes?

Would I find it in the PDF? Partial means right topic, but mixed up.

Pass Partial Fail

3

Does it show a real source?

A real file and page. A made-up file name doesn't count.

Yes No N/A

Then I improve it: Change one thing → Run the test → Score every answer → Fix what failed ↻

EVALUATION

What my first test run found

5 of 6

questions outside my notes got “not covered”

7 of 8

questions my notes cover got an answer, not a refusal

Small test, real method: the same checks work for 14 questions or 1,400.

What testing caught

Mixed up TAM and SAM
Right topic, wrong detail

Invented a source
Cited a file that doesn't exist

A messy refusal
Wrote an essay, then said “not covered”

My own mistake
One test label was wrong, so I fixed it

The verdict: finding the right notes works well. The small free AI writer is what needs upgrading.

WHAT'S NEXT FOR THE PRODUCT

From laptop prototype to company scale

Impact today: one trusted place to revise. Tomorrow: the same design, at company scale.

	TODAY: MY LAPTOP	AT SCALE: WITH API KEYS
The AI writer	Small free model on my laptop	Stronger model through an API key
The notes	14 PM PDFs	Thousands of team documents, with access control
The search memory	Stored on my laptop	A managed cloud database
The testing	14 questions, checked by hand	Hundreds of questions, checked automatically
Cost and privacy	$0, nothing leaves my laptop	Pay per use, with company data agreements

Same guardrails, same tests. Only the parts get bigger.

NEXT IN THE AI LITERACY SERIES

AI evals: how do you know an AI answer is good?

I'll go deeper into test questions, checking answers against sources, and using AI to help grade AI.

- [x] Read about RAG
- [x] Built it
- [x] Tested it
- [ ] AI evals

Riya Murdia